

The background of the slide features an abstract design of overlapping, semi-transparent teal triangles and polygons, creating a complex, crystalline structure that resembles a low-poly landscape or a network diagram. The colors range from light mint green to a deeper teal.

Visually Rich Document Understanding and Intelligence

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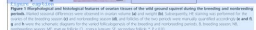


THE UNIVERSITY OF
SYDNEY

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- ***Introduction --- Chapter 1***
 - *Visually Rich Form Document Understanding*
 - *Multi-page Visually Rich Document Question Answering*
 - *Knowledge Transfer for VRD Understanding*
 - *Conclusion and Future Works*

Have you ever tried to analyse the **Visually Rich Documents (VRDs)**, e.g. **Digitalised, Scanned or Handwritten** etc?



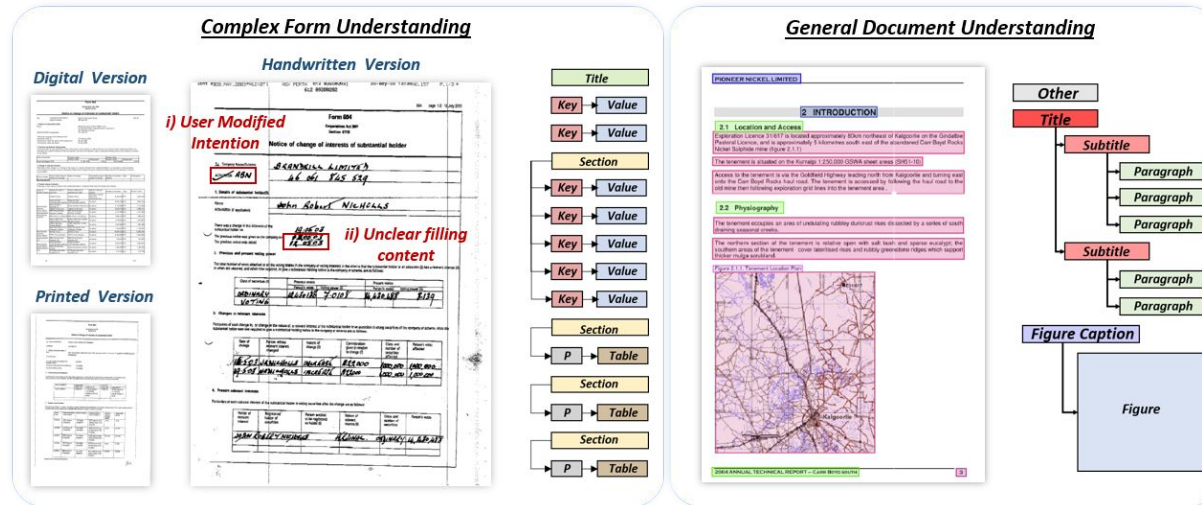
Very tricky problem, especially it requires:

- **Structure Understanding:** understanding layout and/or logical structure of documents.
- **Content Understanding:** extracting the essential information for analysing and assisting decision making.

Introduction

1) Complex Structured VRD Understanding

Some VRDs normally contain complex inherent in structures like forms, involving multiple stakeholders such as designers and users involving **more diversities** and **uncertainties**.



Introduction

2) Multipage VRD QA

Comparing single page scenarios, multi-page is more common but lack of exploration.

Question: Could you show me where the ovarian carcinosarcoma survival nomogram is located and describe what it looks like?

Page 1 **Page 2** **Page 3** **Page 4** **Page 5** **Page 6**

Figure 2

Related Paragraph

Related Paragraph

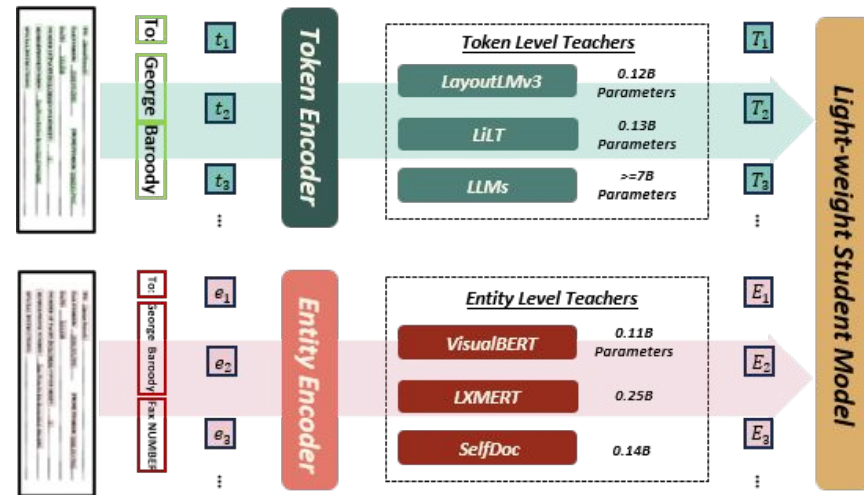
The image displays a multi-page document (Pages 1-6) related to ovarian carcinosarcoma survival nomogram. The document is structured as follows:

- Page 1:** Contains a table of data, likely patient characteristics or survival outcomes, with a red box highlighting a specific section labeled "Related Paragraph".
- Page 2:** Contains a table of data, likely patient characteristics or survival outcomes, with a red box highlighting a specific section labeled "Related Paragraph".
- Page 3:** Contains a table of data, likely patient characteristics or survival outcomes, with a red box highlighting a specific section labeled "Related Paragraph".
- Page 4:** Contains a figure labeled "Figure 2" showing a survival nomogram, which is a graphical representation of survival data, with a red box highlighting the nomogram.
- Page 5:** Contains a figure showing multiple survival curves, likely Kaplan-Meier survival plots, with a red box highlighting a specific section labeled "Related Paragraph".
- Page 6:** Contains a figure showing multiple survival curves, likely Kaplan-Meier survival plots, with a red box highlighting a specific section labeled "Related Paragraph".

Introduction

3) Lack of Exploration in Large Models Integration

A range of large-scale pretrained Vision-Language models are proposed, but lack of related researches focus on distilling knowledge to achieve more representative representations.



Contributions

- **Complex Visually Rich Form Understanding:** Introduces Form-NLU dataset for form structure and key information extraction, with a new model excelling in handling form structures and demonstrating real-world applicability.
- **Multimodal Multi-page VRD QA:** Introduces MMVQA dataset for multimodal entity retrieval in multi-page VRDs, with new frameworks enhancing cross-page retrieving effectiveness.
- **Knowledge Transfer for VRDU:** Presents a multimodal, multi-teacher, multi-task knowledge distillation model to achieve more comprehensive representation and achieving state-of-the-art.

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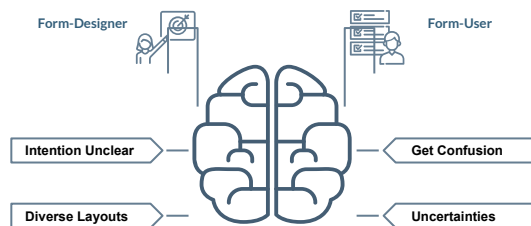
- *Introduction*
- ***Visually Rich Form Document Understanding --- Chapter 3***
Form-NLU: Dataset for the Form Natural Language Understanding.
SIGIR 2023, Rank A*
- *Multi-page Visually Rich Document Question Answering*
- *Knowledge Transfer for VRD Understanding*
- *Conclusion and Future Works*



Visually Rich Form Document Understanding

Above VRD-based problems have been solved successfully especially on some **well-structured document** (e.g. academic papers). However, the form-understanding task is relatively challenging.

- Forms normally involve **multiple stakeholders** normally including Form-Designer and Form-User.
 - Form Designer**: focus on designing a form structure to collect the required information.
 - Form User**: supply the form value of each predefined key by their understanding (**Uncertainty**)



This is a sample form with a header section, a table with multiple columns, and a footer section. It appears to be a form for data collection or reporting.

This is another sample form, similar in structure to the first one, with a header, a table, and a footer. It also appears to be a form for data collection or reporting.

This is a third sample form, similar in structure to the first two, with a header, a table, and a footer. It also appears to be a form for data collection or reporting.

- Forms have a high possibility of having a **combination of various formats**, such **Digital, Printed and Handwritten**. (**Diversity**)

Visually Rich Form Document Understanding

General Intentions

165807110 P

FUNSD
Subset of RVL-CDIP
Scanned only

General Intentions

XFUND
Synthetic Forms

Specific Intentions

(a)

(b)

EPHOIE
Exam Paper Header
Scanned Version

Visually Rich Form Document Understanding

Fine-grained Models

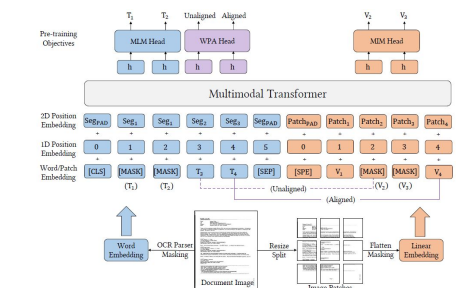


Figure 3: The architecture and pre-training objectives of LayoutLMv3. LayoutLMv3 is a pre-trained multimodal Transformer for Document AI with unified text and image masking objectives. Given an input document image and its corresponding text and layout position information, the model takes the linear projection of patches and word tokens as inputs and encodes them into contextualized vector representations. LayoutLMv3 is pre-trained with discrete token reconstructive objectives of Masked Language Modeling (MLM) and Masked Image Modeling (MIM). Additionally, LayoutLMv3 is pre-trained with a Word-Patch Alignment (WPA) objective to learn cross-modal alignment by predicting whether the corresponding image patch of a text word is masked. "Seg" denotes segment-level positions. "CLS", "MASK", "SEP" and "SPE" are special tokens.

LayoutLMv3
(ACMMM 2022)

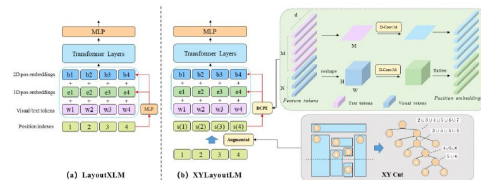


Figure 2: The overview of XYLayoutLM. Different from LayoutLM, our XYLayoutLM proposes Augmented XY Cut and DCPE to extract and utilize layout information for multimodal document understanding. Best viewed in Adobe Acrobat DC.

XYLayoutLM
(CVPR 2022)

Coarse-grained Models

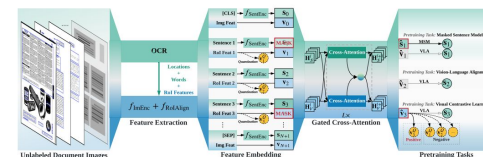


Figure 1: Overview of the proposed approach, UDoc. UDoc first uses a CNN-based visual backbone to learn visual representations. The model then extracts the Region of Interest (RoI) features with OCR bounding boxes and generates a multimodal embedding by combining the textual embedding and position encoding. The transformer-based encoder takes a set of masked multimodal embeddings as input and is pre-trained with three pretraining tasks. All the network parameters except those of the textual encoder are jointly trained during both pretraining and fine-tuning phases.

UDoc
(NIPS 2021)

Visually Rich Form Document Understanding

Contributions

- Introducing **a new dataset** for form structure understanding and key information extraction **to interpret form designer's specific intention** and the alignment of user-written value on it.
- Proposing **a new form understanding model** which **covers the hierarchical structure documents** from words to a semantic entities (e.g. paragraph) and entities to a document page to comprehensively understand the input document pages.
- Extensive qualitative and quantitative analysis are conducted to demonstrate the meaningfulness and effectiveness of proposed datasets and models.

Visually Rich Form Document Understanding

- Dataset Collection**

Form-NLU is a subset of the publicly available financial form data source for Form 604 collected by ASIC (Australian Securities and Investments Commission).

- Task Definition**

Task A: Form Layout Analysis

Form 604
(Companies Act 2006)

Notice of change of interests of substantial holder

2. Company Name/ Scheme: **Tinybeans Group Ltd**

ACN/ARSN: **168 481 614**

1. Details of substantial holder (1)
Name: **Stephen O'Young**
Address: **15/06/2018**

There was a change in the interests of the substantial holder on: **26/04/2017**
The previous notice was given to the company on: **26/04/2017**
The previous notice was dated: **26/04/2017**

3. Previous and present voting power

This table shows the total number of votes attached to all the voting shares in the company or voting interests in the scheme that the substantial holder or an associate (2) held at relevant intervals (3) or when last required, and when now required, to give a substantial holding notice to the company or scheme, are as follows:

Class of securities (4)	Person's votes	Voting power (5)	Person's votes	Voting power (5)
Ordinary securities	6,733,574	6.73%	6,733,574	6.73%

Categories	Description
Title	Form Title (e.g. "Form 604")
Section	Section Header (e.g. "Details of substantial holder")
Form Key	Form Key Segments (e.g. "Name")
Form Value	Form Value Segments (e.g. "Territory Resources Limited")
Table Key	Table Key Segments (e.g. "Voting Power (5)")
Table Value	Table Value Segments (e.g. "5.29%")
Other	Others (e.g. text paragraphs)

Task B: Key Information Extraction

Form 604
(Companies Act 2006)

Notice of change of interests of substantial holder

2. Company Name/ Scheme: **Tinybeans Group Ltd**

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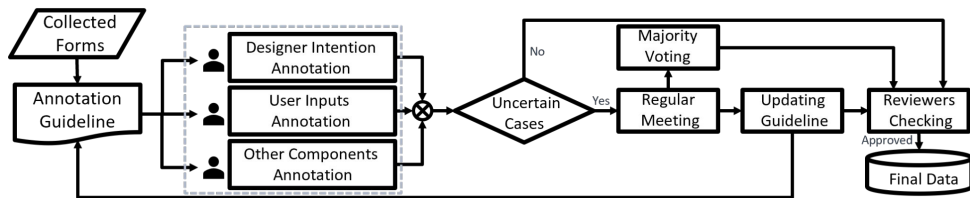
Key	Value
Company Name/ Scheme	Tinybeans Group Ltd
ACN/ARSN	168 481 614
Name	Stephen O'Young
There was a change in the interests of the substantial holder on	15/06/2018
The previous notice was given to the company on	26/04/2017
The previous notice was dated	26/04/2017
Class of securities (4)	Ordinary shares
Previous notice Person's votes	6,733,574
Previous notice Voting power (5)	25.75%
Present notice Person's votes	6,733,574
Present notice Voting power (5)	25.75%

Visually Rich Form Document Understanding

- **Annotation Guidelines**

- Task A: For Layout Analysis we create seven distinct layout component categories including Title, Section, Form Key, Form Value, Table Key, Table Value and Others
- Task B: The Key and Value components from Task A are further distinguished based on the twelve specific Key intentions defined by the domain experts.

- **Annotation Procedures**



- **Human Evaluation**

Two reviewers are invited to check the quality of final version. Cohen's Kappa and Hamming Loss are used to calculate the agreements between two reviewers. We achieve **0.998 Kappa score** and **0.003 Hamming Loss** demonstrating a high annotation quality.

Visually Rich Form Document Understanding

Entity based Statistics

Number of form semantic entities for each data split

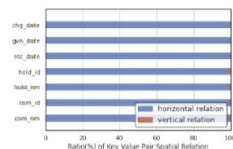
Type	Usage	Form Images	Title	Section	Form		Table		Others
					Key	Value	Key	Value	
Train	Digital	535	1068	1070	3708	3568	2669	2669	1691
		76	152	152	524	510	380	379	246
Test	Digital	146	292	292	1009	978	730	730	458
	Printed	50	98	100	346	332	250	249	152
	Handwritten	50	100	100	348	315	249	226	149
Total Number		857	1710	1714	5935	5703	4278	4253	2696

Bounding Box Characteristics (Geometry and Token Number)

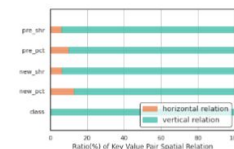
Type	Title	Section	Form		Table	
			Key	Value	Key	Value
Bounding Box Average Width	186.15	136.82	95.46	85.98	57.29	41.92
Bounding Box Average Height	30.44	12.47	12.47	12.47	10.50	10.45
Bounding Box Average PX	4275.94	1720.62	1204.96	1183.10	609.75	456.94
Average Number of Tokens	7.35	7.10	5.16	4.14	4.29	1.74

Key-value Pairs

Spatial Distributions

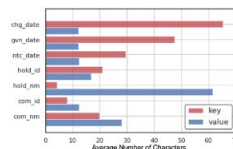


(a) Form Key Value Pairs

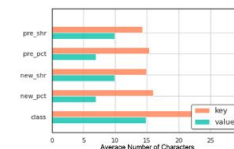


(b) Table Key Value Pairs

Average Number of Characters

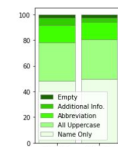


(a) Form Key Value Pairs

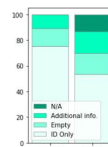


(b) Table Key Value Pairs

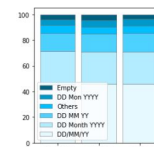
Value Pattern Distributions (User Uncertainties)



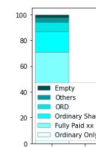
(a) name



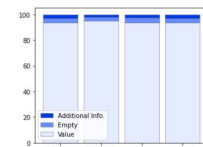
(b) id



(c) dates



(d) class



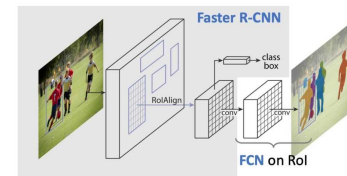
(e) percent/share

Visually Rich Form Document Understanding

• Task A: Form Layout Analysis

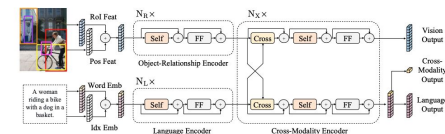
- **Baseline models:** *Faster-RCNN* and *Mask-RCNN* with various backbones (ResNet-50/101)
- **Evaluation Metrics:** we apply **mAP** (mean Average Precision) to evaluate the object detection model performance which is calculated by averaging APs over all per-defined IoU thresholds of all categories.

Mask R-CNN = Faster R-CNN + FCN on ROIs



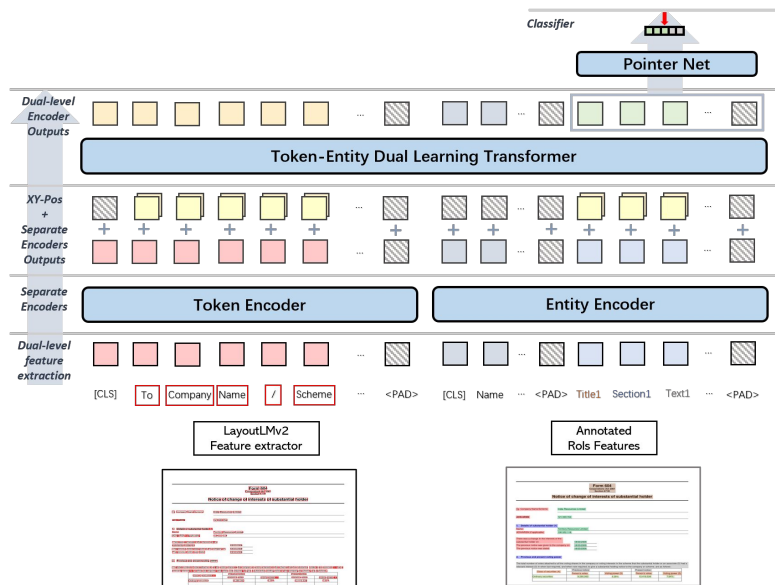
• Task B: Key Information Extraction

- **Baseline models:** some Multi-modal transformer based frameworks are adopted as baselines which can accept RoI feature as inputs including **VisualBERT**, **LXMERT**, **M4C**.
- **Evaluation Metrics:** we use weighted **F1-score** as the primary evaluation metrics for representing overall and breakdown performance.



Visually Rich Form Document Understanding

Joint-grained Form Key Information Extraction



Entity Initial Representation

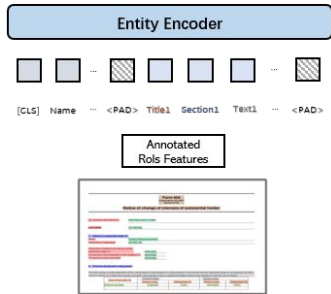
- **Visual:** represent appearance features of input form. ResNet-101 used as backbone to extract visual feature of each semantic entities.
- **Textual:** represent the semantic information of input form. BERT [CLS] token is extracted to represent the textual features of document component.
- **Positional:** positional and spatial distribution of each entity. Different positional encoding methods are adopted for comparing including a new XY-Pos embedding.
- **Gap Distance:** aims to represent the spatial relationship between neighbouring entities
- **Density:** text density features of each entity.

Visually Rich Form Document Understanding

- Entity Encoder**

For **learning semantic relations between entities and key text**.

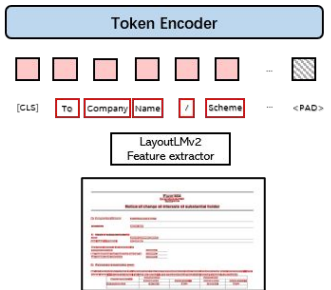
Different pre-trained VLPM with multi-aspect RoI features used to encode the entity-level information.



- Token Encoder**

For acquiring **cross-modal token representations**. LayoutLMv2

used to encode the token-level information.



- XY-Pos Dual Encoder:** is designed for learning the **geometrically sensitive multi-grained** and **multimodality feature representations**.

- XY-Pos: a new geometric-sensitive positional encoding by flattening the stacked sets of normalized bounding box coordinates along with X or Y axis.

$$pos_i^x = [\frac{x_i + d_{i1}}{W_j}, \frac{x_i + d_{i2}}{W_j}, ..., \frac{x_i + d_{im}}{W_j}]$$

- Sum up with both token and entity level XYpos to get final input representations and feed into **Encoder_{dual}**

$$E_{pos_{xy}} = E + X_{entity}^{pos} + Y_{entity}^{pos}$$

$$T_{pos_{xy}} = T + X_{token}^{pos} + Y_{token}^{pos}$$

$$Input_{dual} = concat[T_{pos_{xy}}, E_{pos_{xy}}]$$

$$T_{dual}, E_{dual} = Encoder_{dual}(Input_{dual})$$

- Pointer Net based Classifier**

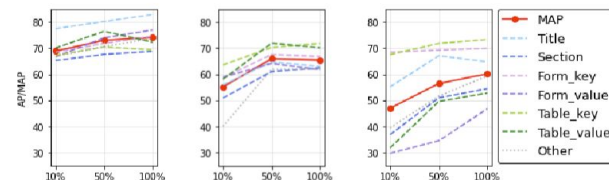
Visually Rich Form Document Understanding

- Overall Performance and Breakdown Results for Form Layout Analysing

Test set	Model	mAP	Breakdowns (Precision)						
			Title	Section	Form_key	Form_value	Table_key	Table_value	Others
$\mathcal{D}$	F-50	68.98	73.54	69.36	67.93	70.37	69.18	66.76	65.73
	F-101	69.99	71.32	68.30	72.37	71.70	68.74	72.84	64.65
	M-50	71.74	77.29	69.06	73.01	70.30	68.40	73.25	70.90
	M-101	71.93	79.54	69.64	71.25	72.38	67.02	74.38	69.33
$\mathcal{P}$	F-50	56.46	57.44	53.98	54.49	51.46	59.31	62.86	55.67
	F-101	59.54	55.72	52.81	63.13	61.32	59.42	68.11	56.26
	M-50	63.06	59.83	60.98	63.74	61.91	70.75	65.93	58.29
	M-101	65.47	63.05	62.49	66.84	61.80	71.82	70.19	62.10
$\mathcal{H}$	F-50	49.87	58.73	43.85	63.14	25.15	63.48	40.14	54.56
	F-101	50.39	53.92	33.36	65.28	36.68	66.97	41.06	55.43
	M-50	57.99	61.45	48.46	68.16	43.14	71.76	56.95	56.05
	M-101	60.22	64.81	54.46	69.97	46.82	73.34	52.81	59.32

Mask-RCNN outperform Faster-RCNN.
Deeper Backbone is more effective.

- Different Training Ratio Effects on Three Test Sets Digital, Printed and Handwritten



(a) Digital($\mathcal{D}$) (b) Printed($\mathcal{P}$) (c) Handwritten($\mathcal{H}$)

Limited effects on Digital sets but **apparent improvements on Printed and Handwritten sets** with the digital training set increasing.

Visually Rich Form Document Understanding

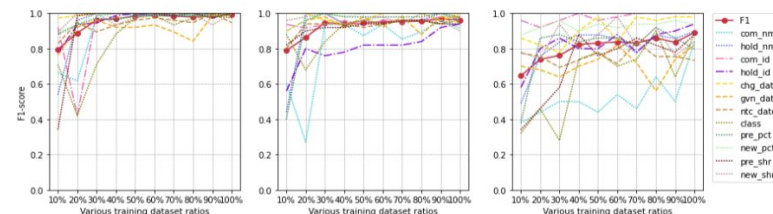
- Overall and Multi-aspect Feature Performance

Model	Input Features					Overall (F1-score)		
	V	T	P	D	G	Digital($\mathcal{D}$)	Printed($\mathcal{P}$)	Handwritten($\mathcal{H}$)
M4C	○	○	○	×	×	96.91	88.62	74.06
	○	○	○	○	×	96.32	89.52	74.98
	○	○	○	×	○	97.00	88.81	75.04
	○	○	○	○	○	97.22	89.78	74.89
VisualBERT	○	×	×	×	×	95.55	85.91	65.25
	○	×	○	×	×	95.84	85.93	67.25
	○	○	○	×	×	96.07	85.90	70.14
	○	○	○	×	○	96.61	87.43	70.79
	○	○	○	○	×	96.70	87.00	71.28
	○	○	○	○	○	96.73	87.18	72.65
LXMERT	○	×	○	×	×	97.83	92.29	80.51
	○	○	○	×	×	97.67	94.15	82.80
	○	○	○	○	×	97.70	94.59	82.60
	○	○	○	×	○	97.74	94.49	83.71
	○	○	○	○	○	97.74	95.07	84.43
Our Model	○	○	○	×	×	99.06	95.07	84.56
	○	○	○	○	×	99.30	95.32	84.75
	○	○	○	×	○	99.12	95.50	85.77
	○	○	○	○	○	99.09	95.50	86.98

- Positional Encoding Validation

PE Approach	Digital($\mathcal{D}$)	Printed($\mathcal{P}$)	Handwritten($\mathcal{H}$)
Without Positional Encoding	98.83	94.34	85.98
Linear Projection	99.09	95.50	86.98
XY-Positional Encoding	99.22	96.14	89.13

- Fine-grained Training Set Ratio

(a) Digital ($\mathcal{D}$)(b) Printed ($\mathcal{P}$)(c) Handwritten ($\mathcal{H}$)

Joint-grained Framework is **effective and robust (Handwritten)**.
Multi-aspect information make more representative features.

Visually Rich Form Document Understanding

Test on PDF Parser Outputs

Form 604
Corporations Act 2001
Section 671B

Notice of change of interests of substantial holder

To Company Name/Scheme: Tinybeans Group Ltd

ACN/ARSN: 168 481 614

1. Details of substantial holder(s)

Name: Stephen O'Young

ACN/ARSN (if applicable): N/A

There was a change in the interests of the substantial holder on: 15/06/2018

The previous notice was given to the company on: 26/04/2017

The previous notice was dated: 26/04/2017

2. Previous and present voting power

The total number of votes attached to all the voting shares in the company or voting interests in the scheme that the substantial holder or an associate (2) had a relevant interest (3) in when the last required, and when now required, to give a substantial holding notice to the company or scheme, are as follows:

Class of securities (4)	Previous notice	Present notice
Person's votes		
Ordinary shares	8,733,574	8,733,574
	25.75%	20.71%

Model	Acc	Breakdowns (Accuracy)											
		nm(name)		id		date			class	pct(percent)		shr(share)	
		com	hold	com	hold	chg	ntc	gvn		pre	new	pre	new
LXMERT	43.78	86.30	80.14	69.18	52.74	72.60	65.07	54.79	4.79	6.85	24.66	5.48	2.74
VisualBERT	57.48	88.36	78.08	63.01	59.59	39.04	70.55	37.67	47.95	50.00	28.77	53.42	73.29
M4C	69.35	83.56	80.82	63.70	73.29	65.75	57.53	82.88	61.64	65.75	58.90	59.59	78.77
Ours	72.72	91.78	81.51	71.92	71.23	80.82	80.82	84.93	71.92	70.55	80.14	50.00	57.53

- *Real-world users are difficult to get the ground truth IoU input. Textline features extracted by PDF parsing tools (PDFMiner) is an alternative way to get the input IoU features.*
- *A transfer learning testing is conducted to test whether our trained model can tackle those application scenarios better.*

Visually Rich Form Document Understanding

- We proposed Form-NLU, **a new form structure understanding and key information extraction dataset**. The proposed dataset covers the important point of view, enabling the interpretation of the form designer's specific intention and the alignment of user-written value on it.
- Moreover, we propose a new model for key information extraction, which applies robust positional and logical relations. Our model outperformed all state-of-the-art models in key-value information extraction tasks.
- We do hope that our proposed dataset and model can be a great insight into form structure and information analysis, hence, we adopted with off-the-shelf pdf layout extraction tool and also provide its feasibility by conducting transfer learning.



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- Introduction
- Visually Rich Form Document Understanding



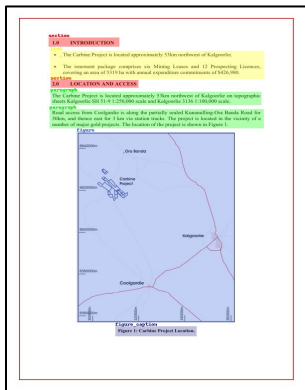
- **Multi-page Visually Rich Document Question Answering --- Chapter 4**

*MMVQA: A Dataset for Multimodal Information Retrieval in PDF-based Visual Question Answering.
IJCAI 2024, Rank A*.*

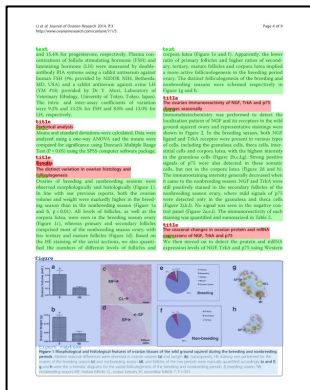
- Knowledge Transfer for VRD Understanding
- Conclusion and Future Works

Multi-page Visually Rich Document Question Answering

- **Document VQA is increasingly evident among several domains**
 - Financial, Mining, Medical, Political and more
- **Answering questions based on documents is knowledge intensive task.**
 - Requires **grounded** and **fact-dependent** approach to acquire **reliable supporting evidence**



Mining Reports



Academic Paper

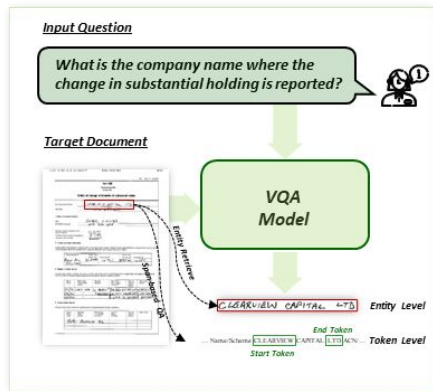


Financial Reports

Sample Document

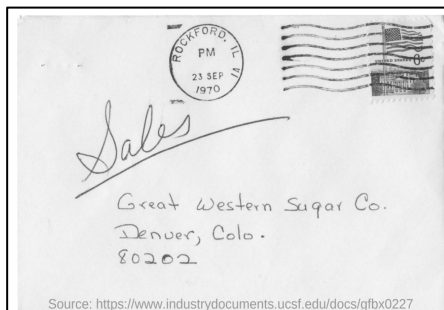


Visual Question Answering



Multi-page Visually Rich Document Question Answering

Single-Page



Q: Mention the ZIP code written?

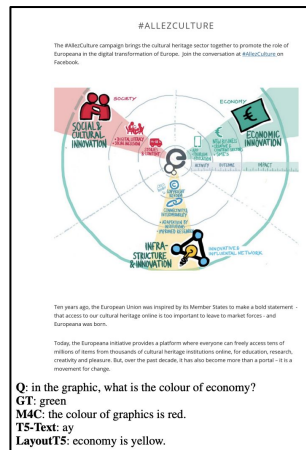
A: 80202

Q: What date is seen on the seal at the top of the letter?

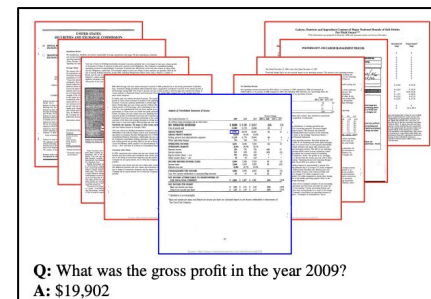
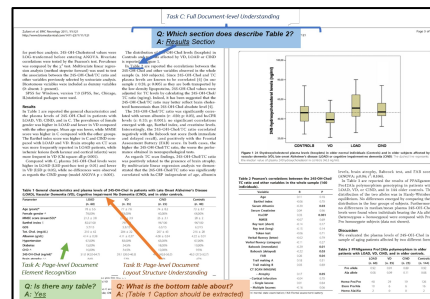
A: 23 sep 1970

Q: Which company address is mentioned on the letter?

A: Great western sugar Co.



Multi-page



DocVQA (WACV 2021)
Industrial Reports
Digital Forms only

VisualMRC (AAAI 2021)
WebPage
Sparse Text

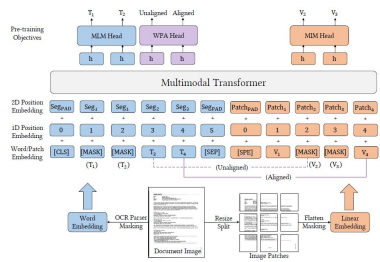
PDF-VQA (ECML-PKDD 2023)
Academic Paper
from PubMed

MP-DocVQA
(Pattern Recognition 2023)

Gaps: All existing Document VQA datasets only focus on *in-line text* or *text-dense entity extraction*, ignoring the *visually-rich entities* such as figure, chart and tables in VRDs.

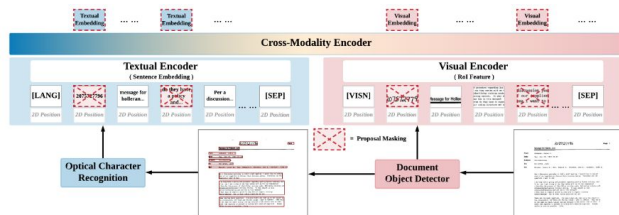
Multi-page Visually Rich Document Question Answering

Token Level Models



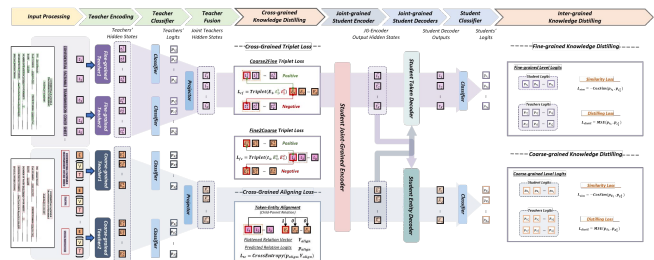
LayoutLMv3
(MM 2022)

Entity Level Models



SelfDoc
(CVPR 2021)

Joint-grained Models



3M VRD
(ACL 2024)

Token-level models are **struggle to capture hierarchical correlations, and limited input length.**

Entity-level models neglect the **fine-grained details, key words and phrases.**

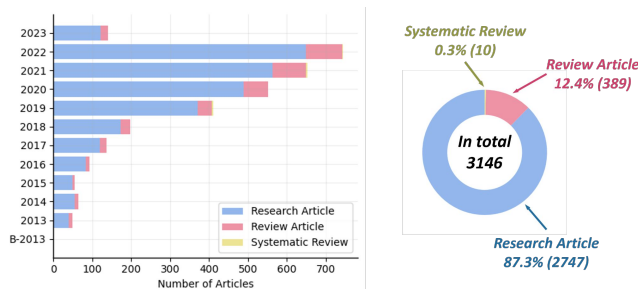
Joint-grained models could mitigate the limitation but are still performing on **single page scenarios.**

Multi-page Visually Rich Document Question Answering

- To introduce **MMVQA**, a new VQA dataset for retrieving **multimodal** document **semantic entities** in **multi-page** VRDs, accompanied by versatile metrics for diverse scenarios.
- A set of frameworks for **multi-page document entity retrieval** is proposed by leveraging the **implicit knowledge** from VLPMs and **fine-grained level information**, enhancing model effectiveness and robustness.
- A series of experiments combining **quantitative** and **qualitative** analyses are performed to provide deeper insights into MMVQA and demonstrate the effectiveness of our proposed techniques for multimodal multi-page document entity retrieval.

Multi-page Visually Rich Document Question Answering

Dataset Source The documents are collected from PubMed Central including PDF and XML formats from [2013](#) to [2023](#).

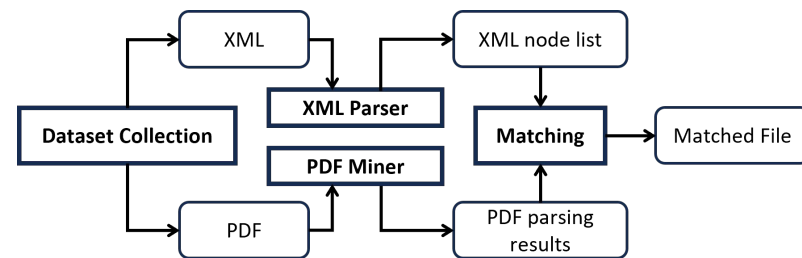


(a) Article Title

(b) Keywords

(c) Abstract

Dataset Preprocessing Following PubLayNet dataset preprocessing workflow by using fuzzy matching to automatically generate **bounding box coordinates** and **assign semantic categories** to document entities.



Semantic entity categories include more detailed types such as **figure caption** and **section**, in addition to basic elements like text and title.

Multi-page Visually Rich Document Question Answering

Source Files

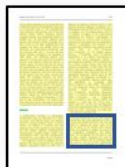
XML Node of Targeting Paragraph

`<p id="Par11">`

SPSS 24.0 (IBM, Chicago, IL, USA) was used for data processing and for all data analyses. Measurement data were expressed in terms of the mean and standard deviation ... set to 0.05. Additionally, prevalence was standardised according to the age structure of the Sixth National Census in 2010.

`</p>`

Document Pages



XML Node of Caption of Targeting Table/Figure

`<table-wrap id="Tab1">`

`<label>Table 1</label>`

`<caption>`

`<p>General characteristics of the participants</p>`

`</caption>`

`<thead>`

...

Paragraph-based Question Generation Procedure

of Questions Setup (N_q)

```
th = sent_tokenizer
sent_list =
th(para_content)
sent_num =
len(sent_list)
If sent_num < 2:
     $N_q = 1$ 
elif sent_num < 4:
     $N_q = 2$ 
Else:
     $N_q = 3$ 
```

Question Generation Prompt

Paragraph: P_c

Ask N_q from that paragraph. The answer to those questions must directly extract sentences from the above paragraph (without any change!).

Output Format:
Q1: XXX



ChatGPT

Generated Question

What software was used for data processing and analysis?

Table/Figure based Question Generation Procedure

Table/Figure Caption Summarisation

Figure (Table) Caption:
figure(table) caption

What does the figure illustrate? (under ten words if possible)

S_c (summarised Caption)

Question Generation Prompt

Question:
Could you find the figure about S_c ?

Improve the previous question by paraphrasing it concisely (shortly and clearly) summarised.



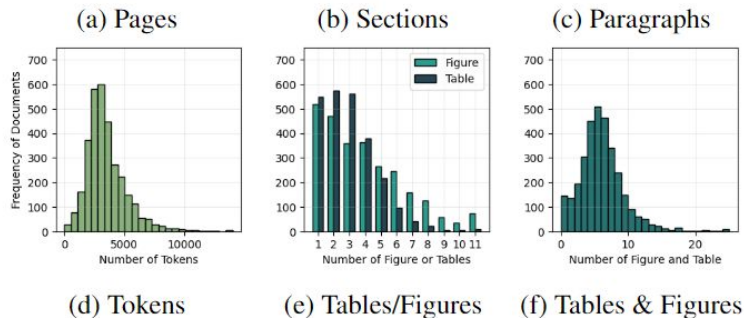
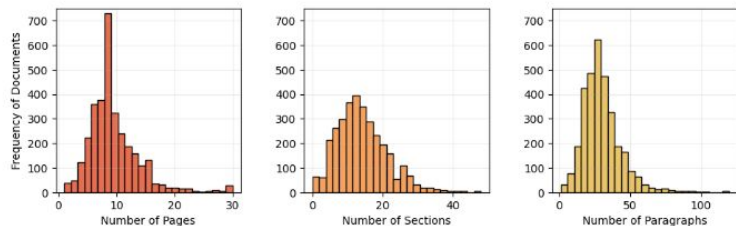
ChatGPT

Generated Question

Can you locate the table displaying the main characteristics of the participants?

Multi-page Visually Rich Document Question Answering

● Component Distributions



Multipage, Lengthy and Visually-Rich

● Super-section based Analysis

We categorise **first-level section** of each document into five **Super-Section**, where the sections under the same Super-Section play similar **structural roles** in a medical domain academic paper, including:

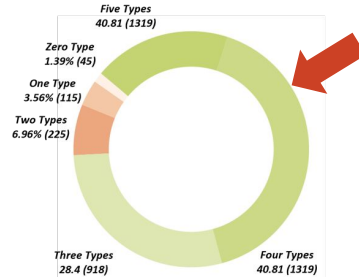
Introduction (Intro), **Material and Method (M&M)**, **Result and Discussion (R&D)**, **Conclusion (Concl)** and **Other**



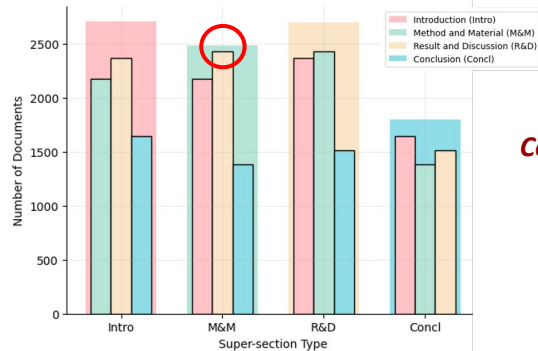
Multi-page Visually Rich Document Question Answering

• Super-Section Correlations and Distributions

Distribution of Super-section Types Across Documents

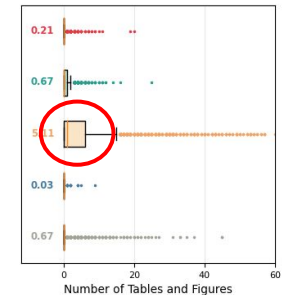
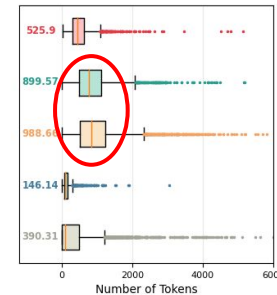
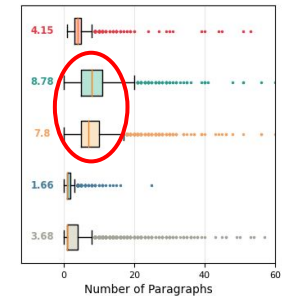
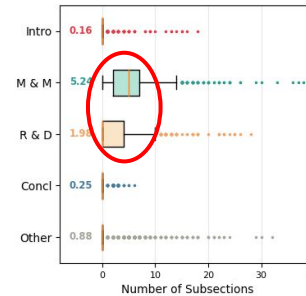


Most of the documents contains more than 4 Super-section types.



**Co-Occurrence examples:
M&M and R&D are highly correlated.**

• Super-Section Component Distribution



- **Question Word Cloud**

- **Super-Section-oriented Question Distribution**



Multi-page Visually Rich Document Question Answering

To propose a **model** for retrieving a set of document entities based on a **question** from **input document images** which are the answer of input question.

Input Document Pages

Page 1 (Q1)



Page 2 (Q1, Q2)



Q1: Paragraph-based Question

What software was used for data processing and analysis?

Targets: (E_{p1}, E_{p2})

E_{p1} E_{p2}

SPSS 14.0 (IBM, Chicago, IL, USA) was used for data processing and for all data analysis. Measurement data were converted in terms of the mean and standard deviation. The test was utilized to compare the mean values of the...

Q2: Table/Figure related Questions

Can you locate the table showing the risk factors for diabetes among adults in Bengbu?

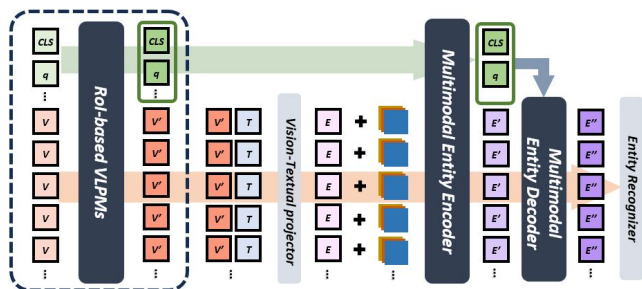
Targets: (E_t)

E_t

Age group	Male	Female	Total
18-24	10	15	25
25-34	20	30	50
35-44	30	40	70
45-54	40	50	90
55-64	50	60	110
65-74	60	70	130
75-84	70	80	150
85-94	80	90	170
95-104	90	100	190
105-114	100	110	210
115-124	110	120	230
125-134	120	130	250
135-144	130	140	270
145-154	140	150	290
155-164	150	160	310
165-174	160	170	330
175-184	170	180	350
185-194	180	190	370
195-204	190	200	390
205-214	200	210	410
215-224	210	220	430
225-234	220	230	450
235-244	230	240	470
245-254	240	250	490
255-264	250	260	510
265-274	260	270	530
275-284	270	280	550
285-294	280	290	570
295-304	290	300	590
305-314	300	310	610
315-324	310	320	630
325-334	320	330	650
335-344	330	340	670
345-354	340	350	690
355-364	350	360	710
365-374	360	370	730
375-384	370	380	750
385-394	380	390	770
395-404	390	400	790
405-414	400	410	810
415-424	410	420	830
425-434	420	430	850
435-444	430	440	870
445-454	440	450	890
455-464	450	460	910
465-474	460	470	930
475-484	470	480	950
485-494	480	490	970
495-504	490	500	990
505-514	500	510	1010
515-524	510	520	1030
525-534	520	530	1050
535-544	530	540	1070
545-554	540	550	1090
555-564	550	560	1110
565-574	560	570	1130
575-584	570	580	1150
585-594	580	590	1170
595-604	590	600	1190
605-614	600	610	1210
615-624	610	620	1230
625-634	620	630	1250
635-644	630	640	1270
645-654	640	650	1290
655-664	650	660	1310
665-674	660	670	1330
675-684	670	680	1350
685-694	680	690	1370
695-704	690	700	1390
705-714	700	710	1410
715-724	710	720	1430
725-734	720	730	1450
735-744	730	740	1470
745-754	740	750	1490
755-764	750	760	1510
765-774	760	770	1530
775-784	770	780	1550
785-794	780	790	1570
795-804	790	800	1590
805-814	800	810	1610
815-824	810	820	1630
825-834	820	830	1650
835-844	830	840	1670
845-854	840	850	1690
855-864	850	860	1710
865-874	860	870	1730
875-884	870	880	1750
885-894	880	890	1770
895-904	890	900	1790
905-914	900	910	1810
915-924	910	920	1830
925-934	920	930	1850
935-944	930	940	1870
945-954	940	950	1890
955-964	950	960	1910
965-974	960	970	1930
975-984	970	980	1950
985-994	980	990	1970
995-1004	990	1000	1990
1005-1014	1000	1010	2010
1015-1024	1010	1020	2030
1025-1034	1020	1030	2050
1035-1044	1030	1040	2070
1045-1054	1040	1050	2090
1055-1064	1050	1060	2110
1065-1074	1060	1070	2130
1075-1084	1070	1080	2150
1085-1094	1080	1090	2170
1095-1104	1090	1100	2190
1105-1114	1100	1110	2210
1115-1124	1110	1120	2230
1125-1134	1120	1130	2250
1135-1144	1130	1140	2270
1145-1154	1140	1150	2290
1155-1164	1150	1160	2310
1165-1174	1160	1170	2330
1175-1184	1170	1180	2350
1185-1194	1180	1190	2370
1195-1204	1190	1200	2390
1205-1214	1200	1210	2410
1215-1224	1210	1220	2430
1225-1234	1220	1230	2450
1235-1244	1230	1240	2470
1245-1254	1240	1250	2490
1255-1264	1250	1260	2510
1265-1274	1260	1270	2530
1275-1284	1270	1280	2550
1285-1294	1280	1290	2570
1295-1304	1290	1300	2590
1305-1314	1300	1310	2610
1315-1324	1310	1320	2630
1325-1334	1320	1330	2650
1335-1344	1330	1340	2670
1345-1354	1340	1350	2690
1355-1364	1350	1360	2710
1365-1374	1360	1370	2730
1375-1384	1370	1380	2750
1385-1394	1380	1390	2770
1395-1404	1390	1400	2790
1405-1414	1400	1410	2810
1415-1424	1410	1420	2830
1425-1434	1420	1430	2850
1435-1444	1430	1440	2870
1445-1454	1440	1450	2890
1455-1464	1450	1460	2910
1465-1474	1460	1470	2930
1475-1484	1470	1480	2950
1485-1494	1480	1490	2970
1495-1504	1490	1500	2990
1505-1514	1500	1510	3010
1515-1524	1510	1520	3030
1525-1534	1520	1530	3050
1535-1544	1530	1540	3070
1545-1554	1540	1550	3090
1555-1564	1550	1560	3110
1565-1574	1560	1570	3130
1575-1584	1570	1580	3150
1585-1594	1580	1590	3170
1595-1604	1590	1600	3190
1605-1614	1600	1610	3210
1615-1624	1610	1620	3230
1625-1634	1620	1630	3250
1635-1644	1630	1640	3270
1645-1654	1640	1650	3290
1655-1664	1650	1660	3310
1665-1674	1660	1670	3330
1675-1684	1670	1680	3350
1685-1694	1680	1690	3370
1695-1704	1690	1700	3390
1705-1714	1700	1710	3410
1715-1724	1710	1720	3430
1725-1734	1720	1730	3450
1735-1744	1730	1740	3470
1745-1754	1740	1750	3490
1755-1764	1750	1760	3510
1765-1774	1760	1770	3530
1775-1784	1770	1780	3550
1785-1794	1780	1790	3570
1795-1804	1790	1800	3590
1805-1814	1800	1810	3610
1815-1824	1810	1820	3630
1825-1834	1820	1830	3650
1835-1844	1830	1840	3670
1845-1854	1840	1850	3690
1855-1864	1850	1860	3710
1865-1874	1860	1870	3730
1875-1884	1870	1880	3750
1885-1894	1880	1890	3770
1895-1904	1890	1900	3790
1905-1914	1900	1910	3810
1915-1924	1910	1920	3830
1925-1934	1920	1930	3850
1935-1944	1930	1940	3870
1945-1954	1940	1950	3890
1955-1964	1950	1960	3910
1965-1974	1960	1970	3930
1975-1984	1970	1980	3950
1985-1994	1980	1990	3970
1995-2004	1990	2000	3990
2005-2014	2000	2010	4010
2015-2024	2010	2020	4030
2025-2034	2020	2030	4050
2035-2044	2030	2040	4070
2045-2054	2040	2050	4090
2055-2064	2050	2060	4110
2065-2074	2060	2070	4130
2075-2084	2070	2080	4150
2085-2094	2080	2090	4170
2095-2104	2090	2100	4190
2105-2114	2100	2110	4210
2115-2124	2110	2120	4230
2125-2134	2120	2130	4250
2135-2144	2130	2140	4270
2145-2154	2140	2150	4290
2155-2164	2150	2160	4310
2165-2174	2160	2170	4330
2175-2184	2170	2180	4350
2185-2194	2180	2190	4370
2195-2204	2190	2200	4390
2205-2214	2200	2210	4410
2215-2224	2210	2220	4430
2225-2234	2220	2230	4450
2235-2244	2230	2240	4470
2245-2254	2240	2250	4490
2255-2264	2250	2260	4510
2265-2274	2260	2270	4530
2275-2284	2270	2280	4550
2285-2294	2280	2290	4570
2295-2304	2290	2300	4590
2305-2314	2300	2310	4610
2315-2324	2310	2320	4630
2325-2334	2320	2330	4650
2335-2344	2330	2340	4670
2345-2354	2340	2350	4690
2355-2364	2350	2360	4710
2365-2374	2360	2370	4730
2375-2384	2370	2380	4750
2385-2394	2380	2390	4770
2395-2404	2390	2400	4790
2405-2414	2400	2410	4810
2415-2424	2410	2420	4830
2425-2434	2420	2430	4850
2435-2444	2430	2440	4870
2445-2454	2440	2450	4890
2455-2464	2450	2460	4910
2465-2474	2460	2470	4930
2475-2484	2470	2480	4950
2485-2494	2480	2490	4970
2495-2504	2490	2500	4990
2505-2514	2500	2510	5010
2515-2524	2510	2520	5030
2525-2534	2520	2530	5050
2535-2544	2530	2540	5070
2545-2554	2540	2550	5090
2555-2564	2550	2560	5110
2565-2574	2560	2570	5130
2575-2584	2570	2580	5150
2585-2594	2580	2590	5170
2595-2604	2590	2600	5190
2605-2614	2600	2610	5210
2615-2624	2610	2620	5230
2625-2634	2620	2630	5250
2635-2644	2630	2640	5270
2645-2654	2640	2650	5290
2655-2664	2650	2660	5310
2665-2674	2660	2670	5330
2675-2684	2670	2680	5350
2685-2			

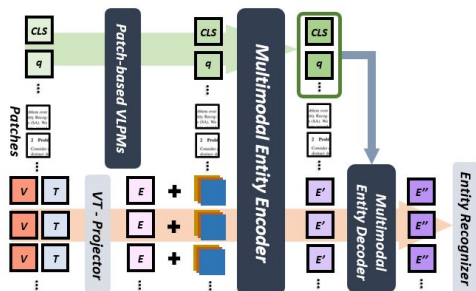
Multi-page Visually Rich Document Question Answering

RoI-based Frameworks



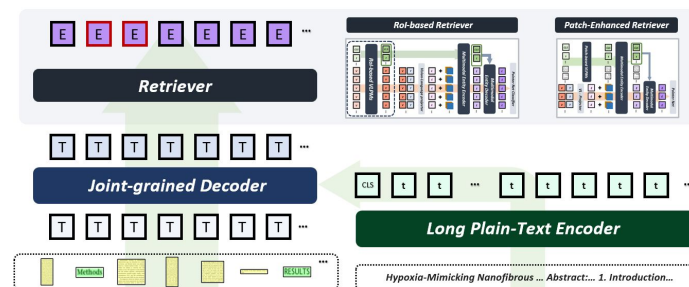
Pretrained RoI-based VLPMs to enhance question and entity (visual) representation

Patch-based Frameworks



Pretrained Patch-based VLPMs to enhance question representation and leverage patch embedding to augment entity features.

Joint-grained Frameworks



Pretrained long-sequence available to acquire **fine-grained representation** to future enhance **entity (textual)** representation.

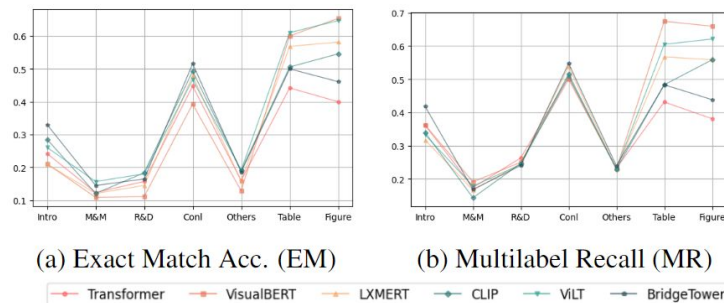
Multi-page Visually Rich Document Question Answering

Overall Performance

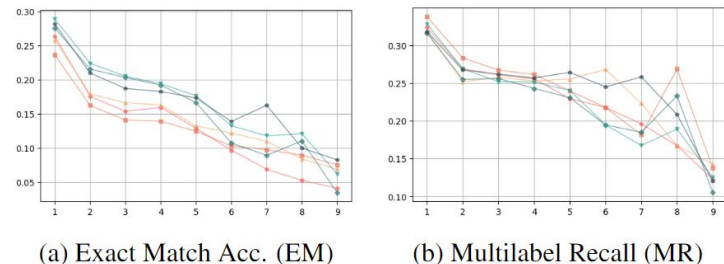
- Patch-based methods outperform RoI-based models on **Exact Matching** and **Partial Matching**.
- RoI-based methods have better **Multi-label Recall** to locate the all correct answer more accurately.

Type	Model	EM		PM		MR	
		Val	Test	Val	Test	Val	Test
RoI-based	Transformer	17.92	19.46	22.48	23.96	25.68	27.50
	VisualBERT	15.39	17.80	21.92	23.86	26.72	28.70
	LXMERT	17.81	19.77	23.37	25.07	25.38	26.86
Patch-based	CLIP	20.71	22.55	25.70	27.59	24.79	26.56
	ViLT	21.71	23.47	27.56	29.14	25.71	27.40
	BridgeTower	19.88	22.37	23.99	26.30	25.37	27.64

Breakdown - Super-Section



Breakdown - Page Range



Multi-page Visually Rich Document Question Answering

Overall Performance

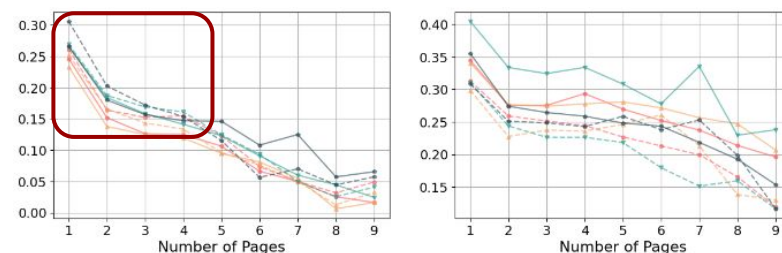
Type	Model	EM		PM		MR	
		Val	Test	Val	Test	Val	Test
RoI-based	Transformer	17.92	19.46	22.48	23.96	25.68	27.50
	VisualBERT	15.39	17.80	21.92	23.86	26.72	28.70
	LXMERT	17.81	19.77	23.37	25.07	25.38	26.86
Patch-based	CLIP	20.71	22.55	25.70	27.59	24.79	26.56
	ViLT	21.71	23.47	27.56	29.14	25.71	27.40
	BridgeTower	19.88	22.37	23.99	26.30	25.37	27.64
Joint-grained BridgeTower	w/ <i>PDFMiner</i>	21.62	23.56	26.63	28.50	27.50	29.22
	w/ <i>OCR</i>	21.53	23.25	26.90	28.56	26.75	28.45

Text-dense Super-Section Breakdown

Model	Overall	Intro	M&M	R&D	Conl	Other
Transformer	17.32	24.19	12.36	15.71	44.82	15.97
Jg-Transformer	18.97	25.14	15.06	17.35	44.38	17.36
LXMERT	16.29	21.00	12.04	14.49	47.95	15.81
Jg-LXMERT	18.33	22.41	15.52	16.53	45.68	17.42
ViLT	19.87	26.06	15.67	18.03	46.76	19.10
Jg-ViLT	20.44	26.36	16.11	19.25	40.93	19.44
BridgeTower	19.95	33.02	14.47	16.46	51.62	18.59
Jg-BridgeTower	22.20	31.47	18.31	19.95	46.98	19.63

Integrating fine-grained level information could benefit text dense entity retrieving.

Page-Range based Breakdown



(a) Exact Match Acc. (EM)

(b) Multilabel Recall (MR)



Applying in Real-world Scenarios

Model	Overall	Intro	M&M	R&D	Conl	Other	Table	Figure
Vanilla BridgeTower	22.37	33.02	14.47	16.46	51.62	18.59	50.03	46.15
Jg-BridgeTower	22.20*	31.47	18.31	19.95	46.98	19.63	N/A	N/A
Jg-BridgeTower- <i>PDFMiner</i>	23.56	31.94	15.80	19.11	52.59	19.10	44.93	46.86
Jg-BridgeTower- <i>OCR</i>	23.25	29.50	16.61	17.82	51.08	17.68	55.07	53.14

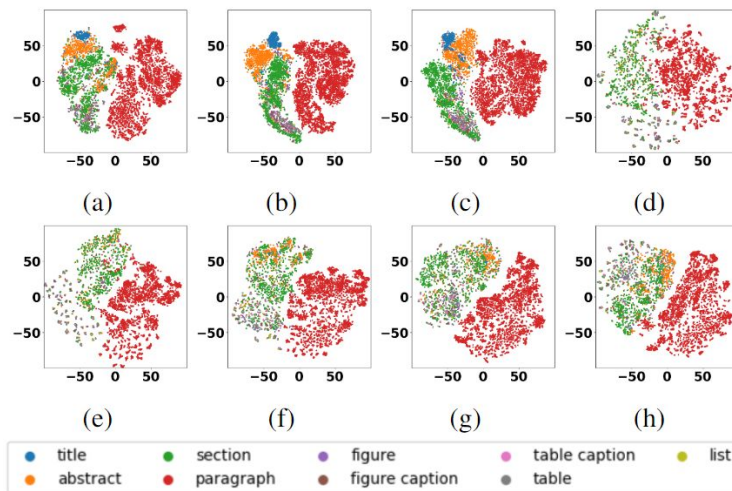
OCR extracts text from figures and tables.

PDFMiner specializes in high-quality text extraction from documents

Multi-page Visually Rich Document Question Answering

Category-oriented Feature Space

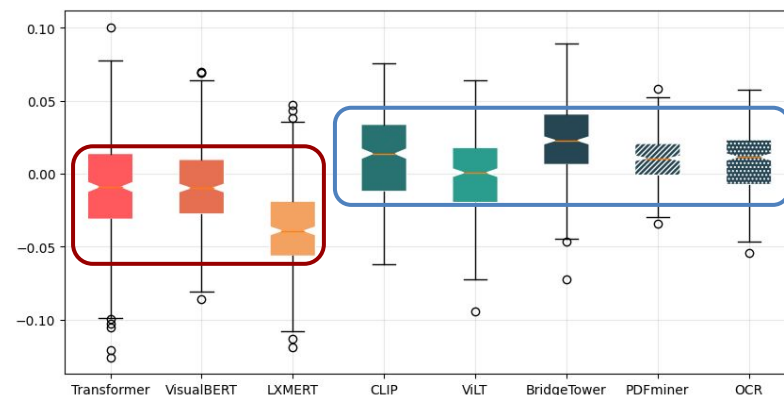
Two-dimensional T-SNE visualisation



Integrating fine-grained information make the text dense entities more distinguishable

Question-Answer Correlation

Average cosine similarity between question [CLS] and averaged target entity embedding.



Patch-based models show better question-answer correlation than RoI-based models in general.

Multi-page Visually Rich Document Question Answering

- Single Page and Multi-page**

Compared with multi-page retrieving, single-page is a easy mode.

Question: What was the role of pyridine-2-aldoxime in the activation process?

Page	Model	Predicts	
	Transformer	{P32}	✓
	VisualBERT	{P32}	✓
	LXMERT	{P32}	✓

	Joint-grained		
	JG-Transformer	{P32}	✓
	JG-LXMERT	{P32}	✓

	Patch-based		
	CLIP	{P32}	✓
	VILT	{P32}	✓
	BridgeTower	{P32}	✓

	Joint-grained		
	JG-VILT	{P32}	✓
	JG-BridgeTower	{P32}	✓

Question: What is the effect of adding lenalidomide or pomalidomide to dexamethasone in patients with renal impairment and RRMM compared to those without renal impairment?

Page 1	Page 2	Page 3	Page 4	Page 5	Model	Predicts	
					Transformer	{P48}	×
					VisualBERT	{}	×
					LXMERT	{}	×

Patch-based							
					CLIP	{P23,24,51}	×
					VILT	{P52}	×
					BridgeTower	{P48,49}	×

Joint-grained							
					PDFminer	{P48,49}	×
					OCR	{P47,48,49}	×

- Text-dense Entity and Visually-rich Entity**

Question: Can you locate the coding scheme table?

Page 1	Page 2	Page 3	Page 4	Page 5	Page 6	Model	Predicts	
						Transformer	{P58}	✓
						VisualBERT	{P58}	✓
						LXMERT	{P58}	✓

Patch-based								
						CLIP	{P58}	✓
						VILT	{P58}	✓
						BridgeTower	{P58}	✓

Joint-grained								
						PDFminer	{P58}	✓
						OCR	{P58}	✓

Compared with text-dense entities (paragraph, list), visually-rich entity retrieving is easier under multi-page setting.

Multi-page Visually Rich Document Question Answering

- *This paper presents a contribution by introducing the **MMVQA dataset** and a **novel joint-grained architecture** for information retrieval from multi-page VRDs.*
 - *The MMVQA from PubMed Central showcases diverse document types, complex structures, and extensive content-related questions in multi-page documents.*
 - *We also introduce the strong benchmark, Joint-grained retrieval architecture, which consistently enhances model performance, particularly in complex document sections.*
- *We hope this research could not only advance the understanding of multi-page document comprehension but also set a foundation for future exploration and refinement of models in this domain, marking a significant step forward in document understanding research.*

Table of Contents

- *Introduction*
- *Visually Rich Form Document Understanding*
- *Multi-page Visually Rich Document Question Answering*



ACL 2024

Bangkok, Thailand

- ***Knowledge Transfer for VRD Understanding --- Chapter 5***

3MVRD: Multimodal Multi-task Multi-teacher Visually-Rich Form Document Understanding
ACL 2024, Rank A.*

- *Conclusion and Future Works*

THE UNIVERSITY OF
SYDNEY

Knowledge Transfer for VRD Understanding

- **Increasing demands of VRDU across several domains.**
- Comparing other document collection, forms involves more challenges.
 - Forms normally involve **at least two parties**: Form-Designer and Form-User.
 - **Form Designer**: focus on developing a form structure to collect the required information by the clear key point.
 - **Form User**: supply the form value of each predefined key by their understanding
 - Forms have a high possibility of having a combination of various natures, such Digital, Scanned-Printed and Scanned-Handwritten.



Academic Papers
PublayNet ICDAR 2019

This image shows a form from the FormNLU SIGIR 2023 dataset. It is a digital form with a table and text. The table has columns for 'Name', 'Age', 'Gender', 'Marital Status', 'Occupation', 'Education', 'Income', 'Religion', 'Nationality', 'Language', 'Address', 'Phone', 'Email', and 'Comments'. The form is filled with handwritten text.

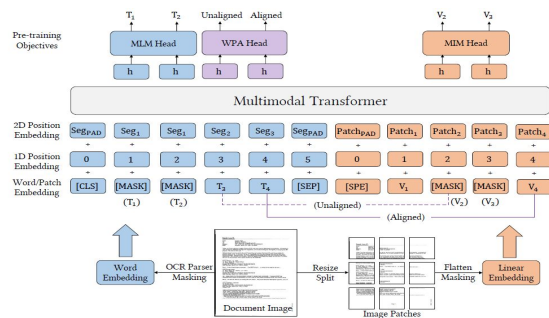
Forms (Digital, Printed, Handwritten)
FormNLU SIGIR 2023

This image shows another form from the FormNLU SIGIR 2023 dataset. It is a digital form with a table and text. The table has columns for 'Name', 'Age', 'Gender', 'Marital Status', 'Occupation', 'Education', 'Income', 'Religion', 'Nationality', 'Language', 'Address', 'Phone', 'Email', and 'Comments'. The form is filled with handwritten text.

This image shows a third form from the FormNLU SIGIR 2023 dataset. It is a digital form with a table and text. The table has columns for 'Name', 'Age', 'Gender', 'Marital Status', 'Occupation', 'Education', 'Income', 'Religion', 'Nationality', 'Language', 'Address', 'Phone', 'Email', and 'Comments'. The form is filled with handwritten text.

Knowledge Transfer for VRD Understanding

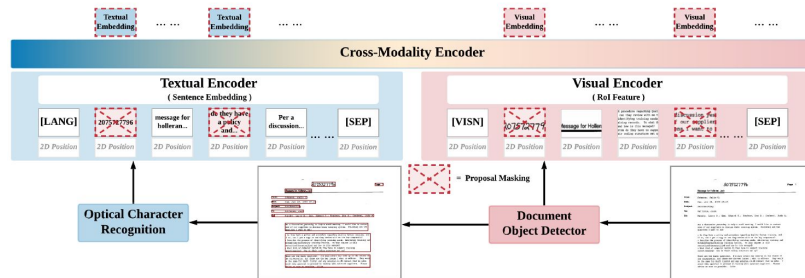
Fine-grained Pretrained VRDU Models



LayoutLMv3, ACMMM 2023

Token-level models are **struggle to capture hierarchical correlations**, and **limited input length**.

Coarse-grained Pretrained VRDU Models



SelfDoc, CVPR 2021

Entity-level models neglect the **fine-grained details**, key words and phrases.

Knowledge Transfer for VRD Understanding

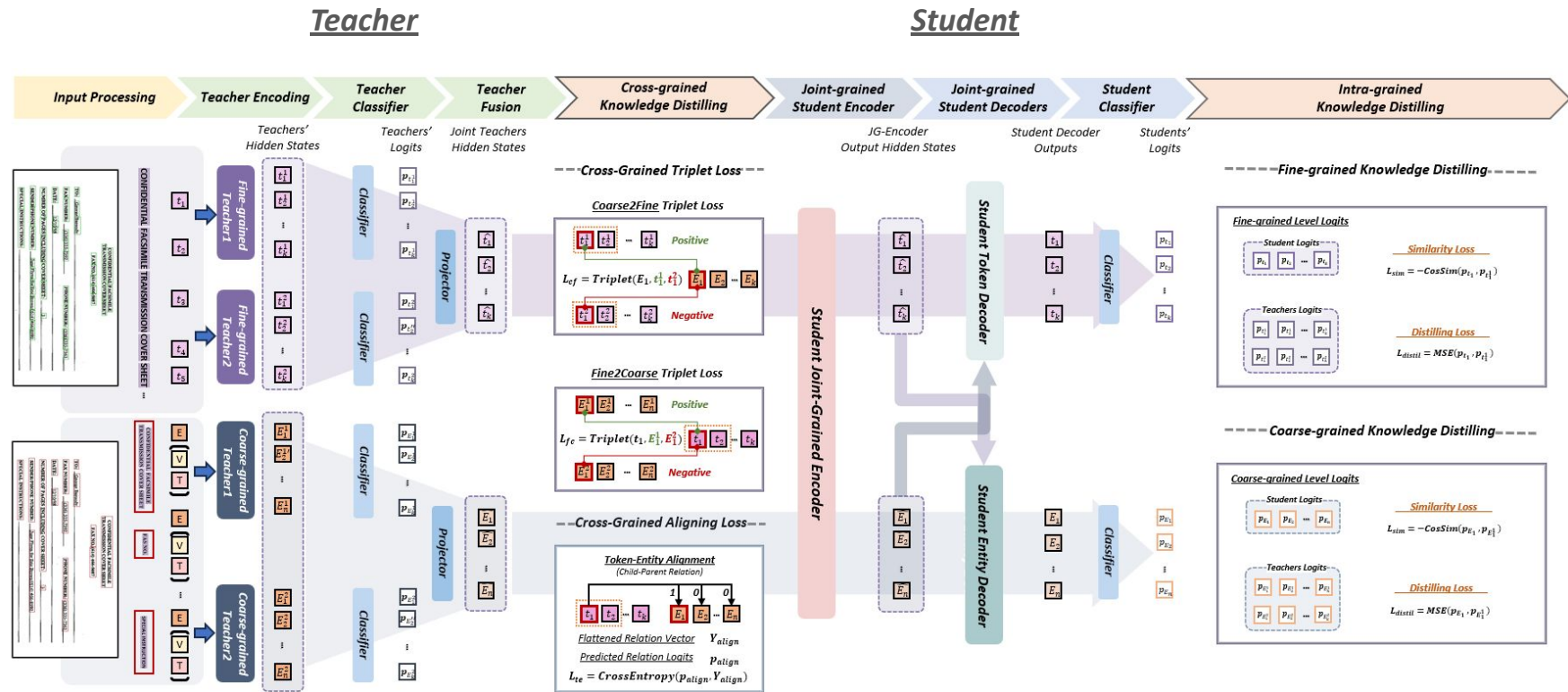
- **Research Aims**

To introduce a novel **joint-grained framework** utilizing **knowledge distillation approaches** to integrate knowledge from **pretrained VRDU models or VLPs** to acquire more comprehensive document representations.

- **Contributions**

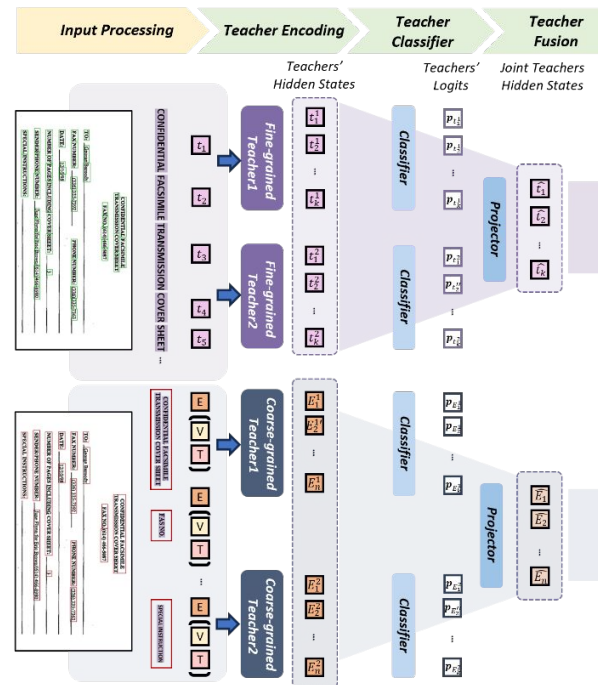
- We present a cutting-edge **Multimodal, Multi-task knowledge distillation** approach specifically designed for understanding visually-rich form documents.
- This research is **the first model** to employ **Multi-teacher knowledge distillation** to integrate different multimodal pretrained models in **form document understanding**.
- Our model demonstrates superior performance on publicly available form document datasets.

Knowledge Transfer for VRD Understanding



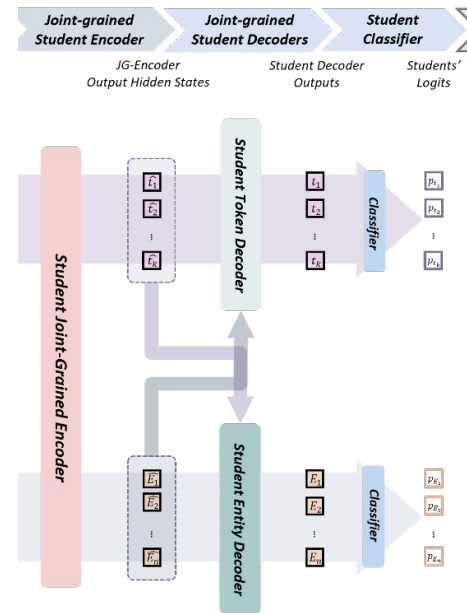
Knowledge Transfer for VRD Understanding

- **Multimodal Multi-Teachers (Fine Tuned)**
 - Fine-Grained Teachers: Sequence Token Tagging
 - Coarse-Grained Teachers: Sequential Entity Classification
- **Inputs:**
 - FG: Token Sequence: $\tilde{\mathbf{t}} = \tilde{t}_1, \tilde{t}_2, \dots, \tilde{t}_k$
 - CG: Entity Embeddings: $\tilde{\mathbf{E}} = \tilde{E}_1, \tilde{E}_2, \dots, \tilde{E}_n$
- **Outputs:**
 - **Hidden States:** $\mathbf{t}^1 = t_1^1, t_2^1, \dots, t_k^1$, $\mathbf{E}^1 = E_1^1, E_2^1, \dots, E_n^1$
 - **Logits:** $\mathbf{p}_{\mathbf{t}^1} = p_{t_1^1}, p_{t_2^1}, \dots, p_{t_k^1}$, $\mathbf{p}_{\mathbf{E}^1} = p_{E_1^1}, p_{E_2^1}, \dots, p_{E_n^1}$
- **Projection**
 - **Combined Representations with Linear Projection**
 $\hat{\mathbf{t}} = \hat{t}_1, \hat{t}_2, \dots, \hat{t}_k$, $\hat{\mathbf{E}} = \hat{E}_1, \hat{E}_2, \dots, \hat{E}_n$



Knowledge Transfer for VRD Understanding

- **Joint-Grained Encoder** $\mathbf{H} = \mathcal{E}(\hat{\mathbf{t}}, \hat{\mathbf{E}})$
 - **Aim:** Learn token-level details and high-level entity structures
 - Captures contextual correlations between fine-grained and coarse-grained representations
- **Joint-Grained Decoders**
 - **Fine-Grained Decoder** $\mathbf{t} = \mathcal{D}_{fg}(\hat{\mathbf{t}}, \hat{\mathbf{E}})$
 - Input: Fine-grained token representations
 - Memory: Coarse-grained entity representations
 - Output: Augmented token representations
 - **Coarse-Grained Decoder** $\mathbf{E} = \mathcal{D}_{cg}(\hat{\mathbf{E}}, \hat{\mathbf{t}})$
 - Input: Coarse-grained entity representations
 - Memory: Fine-grained token representations
 - Output: Comprehensive entity representations



Knowledge Transfer for VRD Understanding

- **Task Oriented Loss:** facilitating a task-based knowledge distillation strategy:

Fine-grained Decoder: $l_t = \text{CrossEntropy}(\mathbf{p}_t, \mathbf{Y}_t)$

Coarse-grained Decoder: $l_e = \text{CrossEntropy}(\mathbf{p}_E, \mathbf{Y}_E)$

- **Intra-grained Loss Functions:** distill valuable information of selected fine-tuned teachers from same granularity.
 - **Similarity loss:** mitigate the logit difference between teacher and student classifier.
 - **Distilling Loss:** to refine the knowledge transfer process, prompting accurate alignment between student and teacher.

- **Cross-grained Loss Functions:** to leverage the inherent correlation between fine or coarse grained features for KD.

- **Cross-grained Triplet Loss - (Teacher Selection):** Leverage the parent-child relation to automatically select more representative teacher to mitigate the distribution distance.
- **Cross-grained Alignment Loss - (Enrich token-entity representation) :** An auxiliary task focuses on predicting the relations between token and entities providing another layer to refine the joint-grained framework.

Knowledge Transfer for VRD Understanding

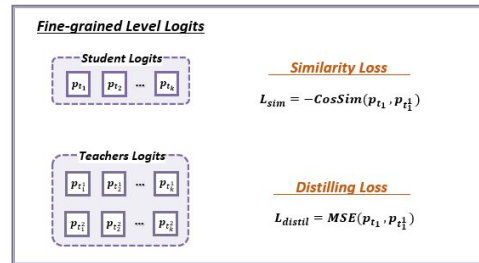
Intra-grained Knowledge Distilling: distill valuable information from selected teachers of same granularity for more robust representation. (Logits)

- **Similarity loss:** mitigate the logit difference between teacher and student classifier.
- Fine-grained:** $l_{sim_t} = - \sum_i \sum_j^{i=n_t, j=k} \text{CosSim}(p_{t_j^i}, p_{t_j})$
- Coarse-grained:** $l_{sim_e} = - \sum_i \sum_j^{i=n_e, j=n} \text{CosSim}(p_{E_j^i}, p_{E_j})$
- **Distilling Loss:** to refine the knowledge transfer process, prompting accurate alignment between student and teacher.

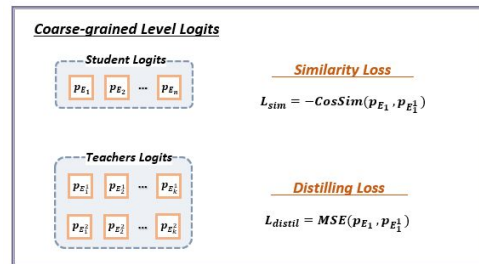
Fine-grained: $l_{distil_t} = \frac{1}{k} \sum_j^{j=k} \text{MSE}(p_{t_j^i}, p_{t_j})$

Coarse-grained: $l_{distil_e} = \frac{1}{n} \sum_j^{j=n} \text{MSE}(p_{E_j^i}, p_{E_j})$

-----Fine-grained Knowledge Distilling-----



-----Coarse-grained Knowledge Distilling-----



Knowledge Transfer for VRD Understanding

Cross-grained Knowledge Distillation: To leverage the inherent correlation between granularities.

- **Cross-grained Triplet Loss:** Leverage the parent-child relation to automatically select more representative teacher to mitigate the cross-grained distribution gap.

- **Coarse2Fine Triplet Loss:**

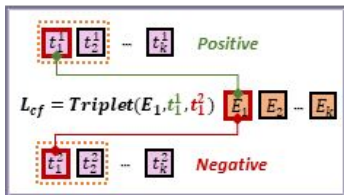
$$l_{\text{triplet}_{fg}} = \frac{1}{k} \sum_i^{i=k} \text{Triplets}(E_j, t_i^{\text{pos}}, t_i^{\text{neg}})$$

- **Fine2Coarse Triplet Loss:**

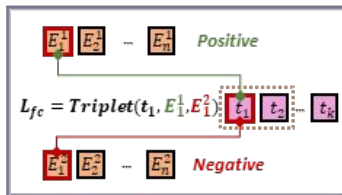
$$l_{\text{triplet}_{cg}} = \frac{1}{k} \sum_i^{i=k} \text{Triplets}(t_i, E_j^{\text{pos}}, E_j^{\text{neg}})$$

---Cross-Grained Triplet Loss---

Coarse2Fine Triplet Loss



Fine2Coarse Triplet Loss

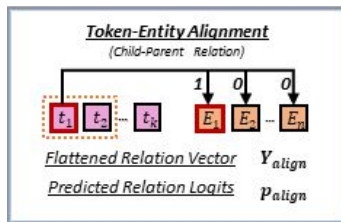


- **Cross-grained Alignment Loss:** An **auxiliary task** focuses on predicting the relations between token and entities to enrich both representations.

An additional layer of refinement to the joint-grained framework. The task is defined as:

$$l_{\text{align}} = \text{CrossEntropy}(\mathbf{p}_{\text{align}}, \mathbf{Y}_{\text{align}})$$

---Cross-Grained Aligning Loss---



Knowledge Transfer for VRD Understanding

FUNSD (ICDAR workshop, 2019), English

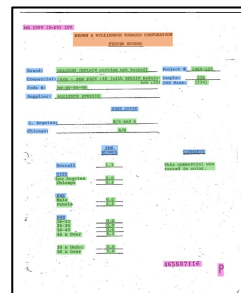
- **Source:** 199 noisy scanned documents from diverse domains
- **Format:** Printed and handwritten, low resolution
- **Evaluation Task:** Sequence token labeling
 - Labels: Question, Answer, Header, Other

FUNSD (Testing)	Question	Answer	Header	Other	Total
Entity	1077	821	122	312	2332
Token	2654	3294	374	2385	8707

FormNLU (Testing)	Title	Section	Form Key	Form Value	Table Key	Table Value	Others	Total
P Entity	98	100	346	332	250	249	152	1527
H Entity	100	100	348	315	249	226	149	1487
P Token	700	1258	1934	1557	993	389	3321	10152
H Token	742	1031	1805	866	779	366	2918	8507

FormNLU (SIGIR 2023), English

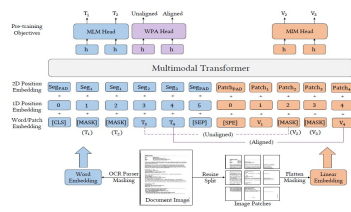
- **Source:** 867 financial form documents from Australian Stock Exchange (ASX) filings
- **Types:** Digital (D), Printed (P), Handwritten (H)
- **Evaluation Task:** Layout analysis
 - Labels: Title, Section, Form Key, Form Value, Table Key, Table Value, Others
 - Focus: Detecting entities in printed (P) and handwritten (H) forms



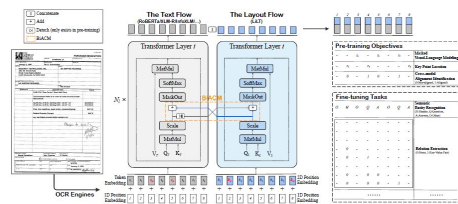
Task A: Form Layout Analysis	
Form Title: [Form Title] Section: [Section Header] Form Key: [Form Key] Form Value: [Form Value] Table Key: [Table Key] Table Value: [Table Value] Other: [Other]	
Categories	Description
Title	Form Title (e.g. "Form 001")
Section	Section Header (e.g. "Details of substantial holder")
Form Key	Form Key Segments (e.g. "Table")
Form Value	Form Value Segments (e.g. "Statutory Provisions Limited")
Table Key	Table Key Segments (e.g. "Voting Power (5)")
Table Value	Table Value Segments (e.g. "5.20%")
Other	Others (e.g. text paragraphs)

- ***Fine-grained Teachers:***

- **LayoutLMv3**: Uses word-image patch alignment with **text**, **layout**, and **image**.
- **LiLT**: Utilizes bi-directional attention between **textual and layout** embeddings, without visual information.
- **BROS**: Employs **relative spatial encoding** between text blocks, without images.



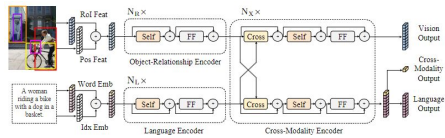
LayoutLMv3, MM 2022



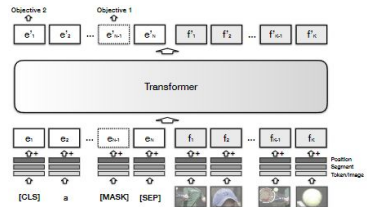
LiLT, ACL 2022

- **Coarse-grained Teachers:**

- **LXMERT**: Uses cross-modality learning between word-level text and object-level image embeddings.
- **VisualBERT**: Inputs image regions and text, relying on implicit alignments without layout information.



LXMERT, EMNLP 2019



VisualBERT, 2019

Knowledge Transfer for VRD Understanding

- **Multi-Teacher Joint-Grained**
 - Improves **FormNLU** printed (96.5% to 97.21%) and handwritten sets (97.39% to 97.75%).
- **Loss Functions:**
 - **Intra-grained** Similarity (Sim) and Distilling (Distil) losses yield higher F1 scores.
 - **Cross-grained** Triplet and alignment (Align) losses outperform baselines on FormNLU.
 - Best performance with combined losses (+Sim+Distil+Triplet+Align).
- **Robustness:**
 - **Intra-grained losses** are **robust** across datasets.
 - Cross-grained losses excel on mono-templated datasets (e.g., FormNLU) but may introduce noise in multi-source datasets (e.g., FUNSD).

Model	Config & Loss	FUNSD	FormNLU	
			<i>P</i>	<i>H</i>
BROS	Single Teacher	82.44	92.45	93.68
LiLT	Single Teacher	87.54	<u>96.50</u>	91.35
LayoutLMv3	Single Teacher	<u>90.61</u>	95.99	<u>97.39</u>
JG- $\mathcal{E}$	Joint Cross Entropy	90.45	94.91	96.55
JG- $\mathcal{D}$	Joint Cross Entropy	90.48	95.68	97.62
JG- $\mathcal{E}$ & $\mathcal{D}$	Joint Cross Entropy	90.57	95.93	97.62
	Joint Cross Entropy	90.53	97.21	97.75
	+ Sim	91.05	98.25	98.09
	+ Distil	90.90	98.12	97.72
MT-JG- $\mathcal{E}$ & $\mathcal{D}$ (Ours)	+ Triplet	90.28	97.58	97.28
	+ Align	90.55	97.24	97.42
	+ Sim + Distil	90.92	98.69	98.39
	+ Triplet + Align			

Knowledge Transfer for VRD Understanding

Multi-Teacher Combination

- **Fine-Grained:** **LayoutLMv3** outperforms LiLT on FUNSD (+3%) and FormNLU (+5%).
- **Coarse-Grained:** **Pre-trained backbones** more robust than randomly initialised Transformers.
- **Multiple Teachers:** Effectively enhance robustness by capturing implicit knowledge.

Effect of Loss Functions

- **Intra-Grained vs. Cross-Grained:** Intra-grained loss more effective alone; combining both enhances distillation.
- **Cross-Grained Losses:** Ineffective alone; effective when combined with intra-grained loss.
- **Stacking Loss Functions:** **Best or second-best performance across all tests.**

FG Teacher	CG Teacher	FUNSD	FormNLU	
			P	H
LLmv3	VBERT	90.19	94.72	96.99
	LXMERT	90.57	95.93	97.62
	Transformer	90.22	93.65	95.94
LiLT	VBERT	87.66	97.65	90.53
	LXMERT	87.34	96.76	91.18
	Transformer	87.91	97.20	90.58
LLmv3	VBERT&LXMERT	90.42	95.05	97.25
LLmv3 & LiLT	LXMERT	90.39	96.73	97.42
LLmv3&LiLT	VBERT&LXMERT	90.53	97.21	97.75

Loss Functions				FUNSD	FormNLU	
Similarity	Distilling	Triplet	Alignment		P	H
O	X	X	X	91.05	98.25	98.09
X	O	X	X	90.90	98.12	97.72
X	X	O	X	90.28	97.58	97.28
X	X	X	O	90.55	97.24	97.42
O	O	X	X	90.63	98.53	97.22
O	X	O	X	90.51	97.71	97.79
O	X	X	O	90.82	97.80	98.05
X	O	O	X	90.82	98.22	98.35
X	O	X	O	90.83	98.63	97.45
O	O	O	X	90.79	98.56	97.72
O	O	X	O	90.66	98.72	97.85
O	O	O	O	90.92	98.69	98.39

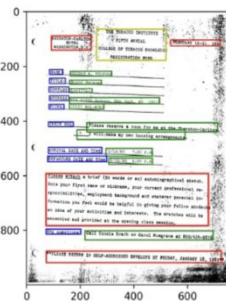
Knowledge Transfer for VRD Understanding

FUNSD Case Study

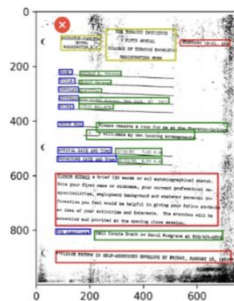
- **Results:** LayoutLMv3 and JG-E&D misrecognized Other.
- **Our Model:** Accurately recognized all document tokens and components.

FormNLU Case Study

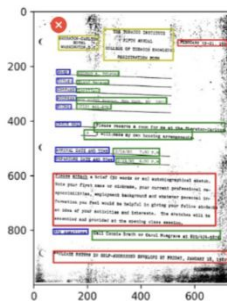
- **Discrepancy:** LayoutLMv3 misidentified **Table Key** as **Form Key**; our model accurately recognized all components.
- **Framework Advantage:** Multi-modal Multi-task Multi-Teacher framework enhances robustness and accuracy by leveraging insights from multiple teachers.



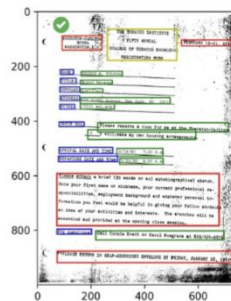
(a) Ground Truth



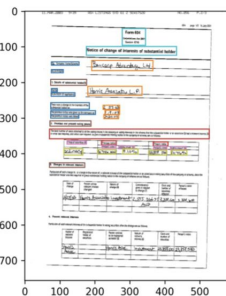
(b) JG-E&D



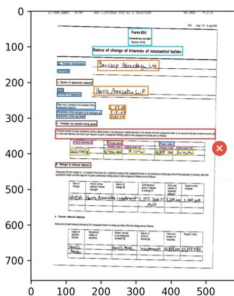
(c) LayoutLMv3



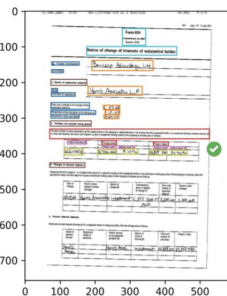
(d) Ours



(a) Ground Truth



(b) LayoutLMv3



(c) Ours

Knowledge Transfer for VRD Understanding

- **Conclusion**

- *Introduced a Multimodal Multi-task Multi-Teacher framework for Visually-Rich Form Documents.*
- *Results show robustness in capturing implicit knowledge from multiple teachers.*
- *Effective in understanding diverse form document natures (scanned, printed, handwritten).*
- *Provides valuable insights into leveraging multi-teacher and multi-loss strategies for document understanding research.*

- **Limitations**

- *Benchmark Scope: Limited high-quality datasets; English-based; lacks diversity/complexity.*
- *Document Understanding Teachers: Relies on general models; lacks form-specific pre-trained models; needs dedicated models.*

Table of Contents

- *Introduction*
- *Visually Rich Form Document Understanding*
- *Multi-page Visually Rich Document Question Answering*
- *Knowledge Transfer for VRD Understanding*
- ***Conclusion and Future Works --- Chapter 6***

Conclusion and Future Works

Conclusion

- This thesis explores the evolving landscape of intelligent understanding of Visually Rich Documents (VRD), highlighting proposing advanced practical solutions and current limitations.
 - To address **complex form document understanding**, Chapter 3 introduces the Form-NLU dataset, which enhances key information extraction from diverse forms with a novel model.
 - Chapter 4 tackles **multimodal information extraction in multi-page** VRDs with the MM-VQA dataset and an advanced model that taps into implicit knowledge from VLPMS.
 - Finally, Chapter 5 presents the 3MVRD framework, which improves document representation by employing **multimodal multi-teacher multi-loss knowledge distillation**.

Future Works

- **Relational-Aware Answer Locating in VRD QA**
- **Domain Adaptation in VRDU**
- **Integration of Off-the-Shelf Tools for Annotation**

Knowledge Transfer for VRD Understanding

Main Observations on FUNSD Dataset

- Joint-Grained frameworks excel in recognizing Questions and Answers.
- Intra-grained loss functions boost Question recognition.
- Benefits of cross-grained loss functions are not well illustrated on the FUNSD dataset.

Model	Config	Overall	Breakdown		
			Header	Question	Answer
LiLT	Teacher	87.54	55.61	90.20	88.34
LayoutLMv3	Teacher	90.61	66.09	91.60	92.78
JG- $\mathcal{E}$	Joint CE	90.45	64.94	91.70	92.67
JG- $\mathcal{D}$	Joint CE	90.48	64.07	91.58	92.73
JG- $\mathcal{E}\&\mathcal{D}$	Joint CE	90.57	64.66	91.48	92.73
MT-JG- $\mathcal{E}\&\mathcal{D}$	Joint CE	90.53	61.24	92.40	91.75
	Sim	91.05	64.81	92.58	92.46
	Distil	90.90	66.96	92.61	91.97
	Triplet	90.28	62.44	92.00	91.44
	Align	90.55	63.81	91.82	92.29
	+Sim+Distil +Triplet+Align	<u>90.92</u>	64.22	92.54	92.31

Main Observations on FormNLU Dataset

- Joint-grained framework and introduced loss functions improve **Section** and **Title** recognition.
- Baseline models perform poorly on **Table keys** and **values**, especially handwritten sets.

Model	Config	FormNLU Printed Overall and Breakdown							FormNLU Handwritten Overall and Breakdown						
		Overall	Sec	Title	F_K	F_V	T_K	T_V	Overall	Sec	Title	F_K	F_V	T_K	T_V
LiLT	Teacher	96.50	98.32	96.97	98.84	96.62	96.57	93.60	91.35	95.39	99.50	94.81	90.67	84.19	89.81
LayoutLMv3	Teacher	95.99	98.45	97.96	97.97	96.73	92.37	92.98	97.39	99.33	99.01	99.85	98.24	93.95	95.95
JG- $\mathcal{E}$	Joint CE	94.91	99.66	98.99	98.11	95.73	90.14	90.31	96.55	99.33	99.01	99.42	98.56	88.37	94.67
JG- $\mathcal{D}$	Joint CE	95.68	99.66	100.00	98.55	96.45	91.94	91.10	97.62	99.33	99.01	99.85	98.56	93.02	95.98
JG- $\mathcal{E}\&\mathcal{D}$	Joint CE	95.93	99.66	97.96	97.82	97.18	91.97	92.15	97.62	99.33	99.01	99.85	98.40	93.74	95.75
MT-JG- $\mathcal{E}\&\mathcal{D}$	Joint CE	97.21	99.32	98.48	99.57	96.58	97.35	95.06	97.75	97.67	99.50	99.13	97.93	95.55	96.41
	Sim	<u>98.25</u>	99.32	99.49	99.28	97.75	97.96	97.12	<u>98.09</u>	99.00	100.00	99.27	98.25	<u>96.45</u>	96.61
	Distil	98.12	99.32	100.00	99.71	<u>97.90</u>	<u>97.55</u>	96.30	97.72	97.35	100.00	99.13	97.62	95.75	97.07
	Triplet	97.58	99.32	99.49	99.28	97.18	<u>97.55</u>	95.87	97.28	98.00	100.00	98.83	97.31	93.90	96.83
	Align	97.24	99.32	98.48	99.71	96.57	96.13	95.47	97.42	99.33	99.50	99.13	96.85	92.86	<u>97.52</u>
	+Sim+Distil +Triplet+Align	98.69	99.32	100.00	99.71	99.25	97.35	97.12	98.39	98.33	100.00	99.56	98.09	96.94	97.75

Q&A - Reviewer 1

- **Q1:** *How are the loss functions combined in the distillation process? Could you clarify the method used to combine the loss functions in the distillation process?*
 - [ANSWERED – p53]
- **Q2:** *Why were only the FUNSD and Form-NLU datasets selected for the experiments on knowledge transfer for VRD understanding? Given the availability of numerous datasets for VRD understanding, could you explain the rationale behind choosing only the FUNSD and Form-NLU datasets for your experiments? What makes these datasets particularly suitable or representative for the study?*
 - [ANSWERED – p42, 51]

Q&A - Reviewer 2

- *The baseline models' experimental results on tasks using the proposed dataset generally show very high performance. What challenges remain in VRD understanding tasks?*

[ANSWERED – p20, 59]

- *Can these challenges be addressed using the proposed dataset, or will new training data and models be required?*

[ANSWERED – p51, 59]

Thank you!



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Appendix



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Visually Rich Form Document Understanding

COUPON CODE REGISTRATION FORM	
6472 FROM	THOM SMITH
6473 TO	VINCE LOSITO
6474 BY	T. BAYLIES, L. GIORDANO, V. LINDSLEY, M. McGLYNN, S. RAPISARLIDIS, mms
BRANDS(S) APPLICABLE	
6481	WAGGINS - HOT ROD
6475 MEDIA	1
6485 ISSUE FREQUENCY/YEAR	1
6486 SPACE/COLOR	FULL PAGE
6480 COUPON ISSUE DATE	5/19/93
6488 COUPON EXPIRATION DATE	11/30/93
CIRCULATION	
6484	25,461
6500 GEOGRAPHICAL AREA(S)	TOWN, HOUSTON, GALESTON, MINNEAPOLIS, ST. PAUL, DOUTH
6502 COUPON VALUE	5.75 OFF 1 PACK
6504 PACK AND OR CARTON	PACK

FEDERAL DISTRIBUTING CORPORATION	
CIGARETTE SAMPLING PROGRAM	
MARKET:	Boston
PRODUCT:	Old Gold Lights
SAMPLING DATES:	October 3 through October 5
SAMPLING HOURS:	9:30 a.m. - 5:30 p.m.
QUANTITIES:	270,000
NUMBER OF SAMPLING PERSONNEL:	24 per day
NUMBER OF SUPERVISORS:	3

Directly applying the model trained on our dataset to different domain dataset, we can still achieve **115/200 correctly** (98/200 on the second best LXMERT model) predicted question answer pairs.

Conclusion and Future Works

- **Relational-Aware Answer Locating in VRD QA**

Focuses on enhancing VRD-QA systems to consider not only the retrieval of direct answers but also the contextual and relational information surrounding the answer.

- **Domain Adaptation in VRDU**

There is a need to explore effective domain adaptation techniques and the utilization of noisy data to reduce the reliance on extensive manual annotations in VRDU tasks. Leveraging automatically generated annotations from tools like OCR, object detection models, and LLMs/MLLMs can help capture domain-specific knowledge and improve model performance with fewer resources.

- **Integration of Off-the-Shelf Tools for Annotation**

The advancement of deep learning has led to the availability of tools that can automatically generate structural and content annotations. Future work should focus on integrating these off-the-shelf tools to reduce the manual effort required for feature extraction and key information retrieval, thereby improving efficiency in real-world VRDU applications.